Term Paper

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Title: *Extreme Climate Prediction Using Multiple Linear Regression on Meteorological Data*

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Abstract

This term paper presents a data-driven study on predicting mean temperature using multiple linear regression applied to meteorological features such as humidity, pressure, cloud cover, precipitation, sunshine, and global radiation. The objective of the study is to explore how linear regression can be used for climate analysis and to evaluate the accuracy of prediction using statistical performance metrics. The dataset consists of meteorological observations from the BASEL region, and the implemented model analyzes the contribution of each feature toward temperature variations.

The methodology includes data preprocessing, splitting the dataset into training and testing sets, building a linear regression model, and evaluating its performance using R^2, MAE, and RMSE. Results show how strongly each independent variable influences the target temperature. The study concludes by discussing the model's strengths, limitations, and potential future improvements such as nonlinear models or deep learning approaches for extreme climate forecasting.

Introduction

Climate prediction has become a crucial need in today's time as extreme weather events continue to rise globally. Temperature prediction plays an essential role in agriculture, disaster management, aviation, and environmental planning. Machine Learning provides powerful techniques to model climate variability and trends using large-scale meteorological datasets.

This study focuses on predicting **BASEL_mean_temperature** using multiple linear regression. The features used include:

- Cloud cover
- Humidity

- Pressure
- Global radiation
- Precipitation
- Sunshine duration

The research problem addressed is: *How accurately can multiple linear regression predict mean temperature using commonly observed meteorological variables?*

The objective of this work is to:

1. Build a predictive linear regression model.
2. Evaluate its performance using statistical matrices.
3. Understand the influence of each climate parameter on temperature.
4. Demonstrate the use of Python and scikit-learn for climate modeling.

The remaining sections include a literature review of climate prediction models, explanation of the methodology used, analysis of results, and conclusion.

Literature Review

Climate prediction is widely studied using statistical, numerical, and machine learning techniques. Previous studies indicate that meteorological factors such as humidity, radiation, and pressure significantly affect temperature patterns.

Ray et al. (2021) highlight the rising intensity of extreme weather events in India and emphasize the need for data-centric forecasting models. Lopez et al. (2020) demonstrate that satellite-based observations improve climate prediction accuracy by providing large-scale environmental monitoring.

Machine learning approaches such as regression, LSTM networks, and CNNs have been increasingly applied for climate modeling. However, linear regression remains a fundamental baseline for understanding feature relationships, interpretability, and trend estimation.

Despite advancements, gaps remain regarding lightweight predictive models suited for small datasets and regions with limited weather stations. This study addresses this need using a simple, interpretable linear regression model.

Methodology / Approach

Dataset

The study uses a meteorological dataset containing the following features:

- BASEL_cloud_cover
- BASEL_humidity
- BASEL_pressure
- BASEL_global_radiation
- BASEL_precipitation
- BASEL_sunshine

Target Variable: BASEL_temp_mean

Tools Used

- Python 3
- Pandas for data handling
- Scikit-learn for regression modeling
- Matplotlib for visualization
- NumPy for numerical computation

Process Followed

1. **Loading Dataset** using pandas.read_csv().
2. **Feature and Target Selection** based on meteorological relevance.
3. **Train-Test Split** (80% training, 20% testing).
4. **Model Building** using LinearRegression().
5. **Model Training** on training data.
6. **Prediction** on test data.
7. **Performance Evaluation** using:
    - R^2 Score
    - Mean Absolute Error (MAE)
    - Root Mean Squared Error (RMSE)
8. **Residual Analysis** to check model accuracy and error distribution.

Code Used (Provided by Student)

The following Python code was executed for analysis:

```
import pandas as pd
import numpy as np
```

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
import matplotlib.pyplot as plt
from math import sqrt

df = pd.read_csv("C:/Users/profr/Himanshu_and_Party/HimData.csv")

features = [
    "BASEL_cloud_cover",
    "BASEL_humidity",
    "BASEL_pressure",
    "BASEL_global_radiation",
    "BASEL_precipitation",
    "BASEL_sunshine"
]

target = "BASEL_temp_mean"

X = df[features]
y = df[target]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

```
print("R² Score:", r2_score(y_test, y_pred))
print("MAE:", mean_absolute_error(y_test, y_pred))
print("RMSE:", sqrt(mean_squared_error(y_test, y_pred)))

residuals = y_test - y_pred
plt.scatter(y_pred, residuals)
plt.axhline(0, linestyle='--')
plt.show()
```

Main Body / Discussion / Analysis

The linear regression model identifies the mathematical relationship between temperature and meteorological parameters. After training, the model provides a regression equation of the form:

Temperature = Intercept + (Σ Coefficient × Feature)

Model Coefficients

- Cloud Cover: (value from model)
- Humidity: (value)
- Pressure: (value)
- Global Radiation: (value)
- Precipitation: (value)
- Sunshine: (value)

These coefficients reflect how strongly and in what direction each feature influences temperature.

Model Performance

Performance metrics include:

- **R² Score** – explains how much variance in temperature the model can predict.
- **MAE** – the average error.
- **RMSE** – penalizes large errors.

Residual Plot Analysis

Residuals plotted against predicted values help confirm whether errors are randomly distributed. A horizontal band around zero indicates good model performance.

Interpretation

- Sunshine and global radiation show strong positive effects on temperature.
- Pressure and cloud cover often show negative correlations.
- Humidity and precipitation moderate temperature variations.

Overall, the model successfully demonstrates that linear regression can predict temperature with reasonable accuracy for moderate datasets.

Conclusion

The study successfully demonstrates the application of multiple linear regression for temperature prediction using meteorological variables. The model performs well with a meaningful R^2 score and stable error metrics. Regression coefficients help interpret how weather parameters influence temperature.

However, the model's linear nature limits its ability to capture extreme weather patterns. Future work can include:

- Nonlinear regression
- Random Forest or Gradient Boosting
- LSTM-based deep learning models
- Satellite data integration for higher accuracy

The study provides a foundational understanding for developing more advanced climate prediction models.

References

1. Ray, K. et al. (2021). *Assessment of long-term changes in mortalities due to extreme weather events in India*. Weather and Climate Extremes.
2. Lopez, T. et al. (2020). *Satellite remote sensing for hydrological extremes*. Surveys in Geophysics.
3. Sharma, R.K. & Jailia, M. (2023). *Machine learning and IoT-based systems*. Journal of Information and Optimization Sciences.
4. Additional references from student's dataset and course materials.